MATHEMATICS

Analysis and approaches

Standard Level | Set A | Paper 1

Mock examination · 2025 series

Time allowed: 1 hour 30 minutes Maximum mark: 80

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A calculator is not permitted.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 5]

Let $f(x) = \frac{x-1}{2-x}$, with domain $\mathbb{R} \setminus \{2\}$.

(a) State the range of f. [1]

(b) Find an expression for $f^{-1}(x)$. [4]

SECTION A

Question 2 [Maximum mark: 6]

Angles in this question are measured in radians.

(a) Express $8 \sin x \cos x$ as a single sine expression. [2]

(b) Hence evaluate $\int_0^{\pi/2} 8 \sin x \cos x \, \mathrm{d}x$. [4]

SECTION A

Question 3 [Maximum mark: 6]

Angles are measured in degrees.

Solve $\sin(2x - 10^\circ) = 1$ for $0^\circ \leq x \leq 360^\circ$. [6]

SECTION A

Question 4 [Maximum mark: 5]

A discrete random variable X has the following distribution.

x	1	2	3
$P(X = x)$	p	$0.7 - p$	q

The expectation of X is 2.1.

(a) Find q. [2]

(b) Find p. [3]

SECTION A

Question 5 [Maximum mark: 6]

Consider logarithms of real arguments.

(a) Write $\log_8(x+2)$ in terms of $\log_2(x+2)$. [1]

(b) Hence solve $3\log_8(x+2) - 2\log_2(x+1) = 1$. [5]

SECTION A

Question 6 [Maximum mark: 6]

An arithmetic sequence has first term u_1 and common difference d. Its first n terms have sum S_n. It is given that $S_3 = S_8$.

(a) Show that $u_1 = -5d$. [4]

(b) Hence find u_6. [2]

SECTION A

Question 7 [Maximum mark: 6]

Let $f(x) = ax^3 + 2ax^2 + bx + 3a$, where $a, b \in \mathbb{R}$. The graph has a stationary point at $(1, 3)$.

(a) Find $f\prime(x)$. [2]

(b) Find a and b. [4]

SECTION B

Question 8 [Maximum mark: 11]

Let $f(x) = ax^2 + bx + 8$, where $a \neq 0$, and let $g(x) = 4ax^2 + 2bx + 8$ and $h(x) = 4ax^2 + 4bx + 33$.

(a)(i)) Write $g(x)$ in terms of f and describe the transformation taking the graph of f to that of g. [3]

(a)(ii)) Two points on the graph of f have x-coordinates 2 and 4. State the x-coordinates of their corresponding points on the graph of g. [1]

(b) Describe a sequence of transformations taking the graph of f to that of h. [3]

(c) The vertex of the graph of f is $(-1, a)$. Find a and b. [4]

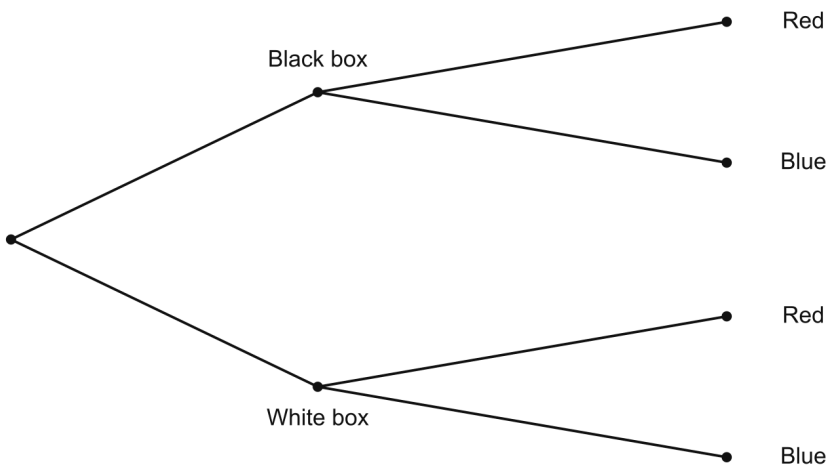
SECTION B

Question 9 [Maximum mark: 14]

A black box contains 1 red ball and 4 blue balls. A white box contains 2 red balls and 3 blue balls. A fair six-sided die is rolled: the black box is selected for scores 1,2,3 and the white box otherwise. One ball is drawn at random from the selected box.

For part(d) only, change the rule: select the black box when the die score is a or below, where a is an integer from 1 to 5; otherwise select the white box.

Selection of a box followed by the colour of a ball



Copy and complete the branch probabilities.

(a) Copy and complete the probability tree, labelling every branch. [3]

(b) Find the probability that the drawn ball is blue. [2]

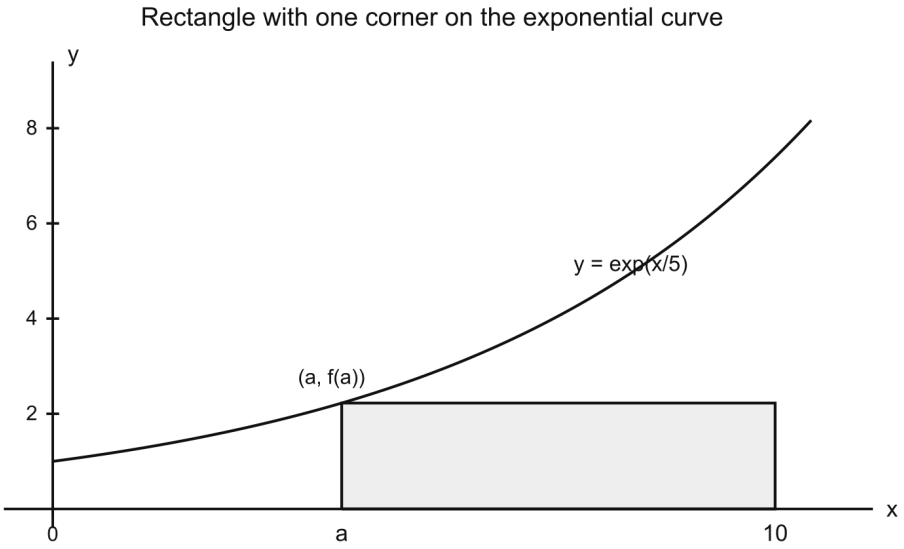
(c) Given that the drawn ball is red, find the probability that it came from the black box. [3]

(d) Under the changed rule, $P(\text{black} \mid \text{red}) = 1/5$. Find a. [6]

SECTION B

Question 10 [Maximum mark: 15]

Let $f(x) = e^{x/5}$. A rectangle has vertices $(a, 0)$, $(10, 0)$, $(10, f(a))$ and $(a, f(a))$, where $0 < a < 10$. Its area is S and perimeter is P.



(a) Find the equation of the tangent to the graph of f at $x = 0$. [3]

(b) Show that $S = (10 - a)e^{a/5}$. [2]

(c) Write an expression for P in terms of a. [2]

(d) Find $\mathrm{d}S/\mathrm{d}a$. [3]

(e) Find the perimeter of the rectangle whose area is greatest. Justify that its area is a maximum. [5]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 5]

(a) Range: $y \neq -1$ **A1**

[1 mark]

(b) $f(x) = y \Leftrightarrow \frac{x-1}{2-x} = y$ **M1**

$\Leftrightarrow x - 1 = y(2 - x)$

$\Leftrightarrow x - 1 = 2y - xy$

$\Leftrightarrow x(1 + y) = 2y + 1$ **M1**

$\Leftrightarrow x = \frac{2y+1}{y+1}$ **A1**

Hence, $\Leftrightarrow f^{-1}(x) = \frac{2x+1}{x+1}$ **A1**

[4 marks]

2. [Maximum mark: 6]

(a) $8\sin x\cos x = 4\times 2\sin x\cos x = 4\sin 2x$ **M1 A1**

[2 marks]

(b) $\int_0^{\frac{\pi}{2}}(8\sin x\cos x)\,dx = \int_0^{\frac{\pi}{2}}(4\sin 2x)\,dx$

$= \left[-2\cos 2x\right]_0^{\pi/2}$ **A1**

$= -2\cos\pi + 2\cos 0$ **M1**

$= 2+2$ **A1**

$= 4$ **A1**

[4 marks]

3. [Maximum mark: 6]

$\sin(2x-10°)=1$, for $0° \le x \le 360°$.

$2x-10°=90°+360°k$ **M1 A1**

$2x=100°+360°k$ **A1**

$x=50°+180°k$ **A1**

Solutions: $x=50°$, $x=230°$ **A1 A1**

4. [Maximum mark: 5]

(a) $p+0.7-p+q=1 \Leftrightarrow 0.7+q=1 \Leftrightarrow q=0.3$ **M1 A1**

[2 marks]

(b) $\mathrm{E}(X)=2.1 \Leftrightarrow p+2(0.7-p)+3q=2.1$ **M1 A1**

$\Leftrightarrow p+1.4-2p+0.9=2.1 \Leftrightarrow p=0.2$ **A1**

[3 marks]

5. [Maximum mark: 6]

(a) Change of base: $\log_8(x+2)=\dfrac{\log_2(x+2)}{\log_2 8}=\dfrac{\log_2(x+2)}{3}$ **A1**

[1 mark]

(b) Therefore,

$3\log_8(x+2)-2\log_2(x+1)=1$

$\Leftrightarrow \log_2(x+2)-2\log_2(x+1)=1$

$\Leftrightarrow \log_2\dfrac{(x+2)}{(x+1)^2}=1$ **M1 A1**

$\Leftrightarrow \dfrac{(x+2)}{(x+1)^2}=2$ **M1**

$\Leftrightarrow x+2=2(x+1)^2 \Leftrightarrow x+2=2(x^2+2x+1)$

$\Leftrightarrow 2x^2+3x=0 \Leftrightarrow x(2x+3)=0$ **A1**

$\Leftrightarrow x=0$ or $x=-\dfrac{3}{2}$ (rejected)

Hence, $x=0$. **A1**

[5 marks]

6. [Maximum mark: 6]

(a) $S_3 = S_8 \Leftrightarrow \frac{3}{2}(2u_1 + 2d) = \frac{8}{2}(2u_1 + 7d)$ **M1 A1 A1**

$\Leftrightarrow 3u_1 + 3d = 8u_1 + 28d$

$\Leftrightarrow 0 = 5u_1 + 25d \Leftrightarrow 5u_1 = -25d$ **A1**

$\Leftrightarrow u_1 = -5d$

[4 marks]

(b) $u_6 = u_1 + 5d = 0$. **M1 A1**

[2 marks]

7. [Maximum mark: 6]

Let $f(x) = ax^3 + 2ax^2 + bx + 3a$

(a) $f'(x) = 3ax^2 + 4ax + b$. **A1 A1**

[2 marks]

(b) $f(1) = 3 \Leftrightarrow a + 2a + b + 3a = 3 \Leftrightarrow 6a + b = 3$ **A1**

$f'(1) = 0 \Leftrightarrow 3a + 4a + b = 0 \Leftrightarrow 7a + b = 0$ **M1 A1**

$a = -3,\ b = 21$ **A1**

[4 marks]

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

8. [Maximum mark: 11]

$f(x) = ax^2 + bx + 8$.

$g(x) = 4ax^2 + 2bx + 8$

(a) (i) $g(x) = f(2x)$: Horizontal stretch with scale factor $\frac{1}{2}$ (i.e. shrink) **A1 A1 A1**

(ii) $x = 1$, $x = 2$ **A1**

[4 marks]

$h(x) = 4ax^2 + 4bx + 33$

(b) $h(x) = 4f(x) + 1$ **M1**

Vertical stretch with scale factor 4 **A1**

Vertical translation 1 unit up. **A1**

[3 marks]

(c) **METHOD A**

$-\frac{b}{2a} = -1 \Leftrightarrow b = 2a$ **M1**

$f(-1) = a \Leftrightarrow a - b + 8 = a \Leftrightarrow b = 8$ **M1 A1**

$a = 4$ **A1**

METHOD B

The vertex form is $f(x) = a(x+1)^2 + a$ **M1**

$= a(x^2 + 2x + 1) + a = ax^2 + 2ax + 2a$

Hence, compared with the original **M1**

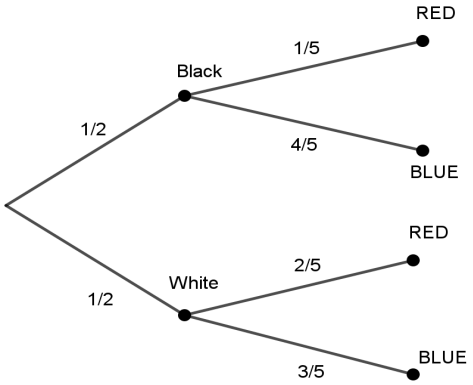
$2a = 8a \Rightarrow a = 4$ **A1**

$b = 2a \Rightarrow b = 8$ **A1**

[4 marks]

9. [Maximum mark: 14]

(a)



A1 A1 A1
[3 marks]

(b) $\mathrm{P}(BLUE) = \frac{1}{2}\times\frac{4}{5}+\frac{1}{2}\times\frac{3}{5} = \frac{4}{10}+\frac{3}{10} = \frac{7}{10}$ **A1 A1**

[2 marks]

(c) $\mathrm{P}(RED) = 1-\frac{7}{10} = \frac{3}{10}$ **OR** $\mathrm{P}(RED) = \frac{1}{2}\times\frac{1}{5}+\frac{1}{2}\times\frac{2}{5} = \frac{1}{10}+\frac{2}{10} = \frac{3}{10}$ **A1**

$\mathrm{P}(\textbf{\textit{Black Box}}/RED) = \frac{\frac{1}{2}\times\frac{1}{5}}{\frac{3}{10}} = \frac{\frac{1}{2}\times\frac{1}{5}}{\frac{3}{10}} = \frac{1}{3}$ **M1 A1**

[3 marks]

(d) Let the probability that the Black box is selected (a or below) be p (OR $p = \frac{a}{6}$) **M1**

$\mathrm{P}(RED) = p\times\frac{1}{5}+(1-p)\times\frac{2}{5} = \frac{p}{5}+\frac{2-2p}{5} = \frac{2-p}{5}$ **A1**

$\mathrm{P}(\textbf{\textit{Black Box}}/RED) = \frac{\frac{p}{5}}{\frac{2-p}{5}} = \frac{p}{2-p}$ **M1 A1**

$\frac{p}{2-p} = \frac{1}{5} \Leftrightarrow 5p = 2-p \Leftrightarrow 6p = 2 \Leftrightarrow p = \frac{2}{6}\ \left(=\frac{1}{3}\right)$ **A1**

Hence, we select the Black box if the die shows 1 or 2
Hence, $a = 2$. **A1**

[6 marks]

10. [Maximum mark: 15]

$f(x)=e^{\frac{x}{5}}$.

(a) At $x=0$, $y=1$.

$f'(x)=\frac{1}{5}e^{\frac{x}{5}}$ **M1 A1**

$m_T=\frac{1}{5}$

Tangent line: $y=\frac{1}{5}x+1$ **A1**

[3 marks]

(b) The base is $10-a$ **A1**

The heigh is $f(a)=e^{\frac{a}{5}}$ **A1**

Hence $S=(10-a)e^{\frac{a}{5}}=10e^{\frac{a}{5}}-ae^{\frac{a}{5}}$

[2 marks]

(c) $P=2(10-a)+2e^{\frac{a}{5}}=20-2a+2e^{\frac{a}{5}}$ **A1 A1**

[2 marks]

(d) $\frac{dS}{da}=2e^{\frac{a}{5}}-\left(e^{\frac{a}{5}}+a\frac{1}{5}e^{\frac{a}{5}}\right)$. **M1 A1**

$=e^{\frac{a}{5}}-\frac{a}{5}e^{\frac{a}{5}}$. **A1**

[3 marks]

(e) $\frac{dS}{da}=0\Leftrightarrow e^{\frac{a}{5}}=\frac{a}{5}e^{\frac{a}{5}}\Leftrightarrow 1=\frac{a}{5}\Leftrightarrow a=5$ **M1 A1**

The second derivative is $\frac{d^2S}{da^2}=\frac{1}{5}e^{\frac{a}{5}}-\left(\frac{1}{5}e^{\frac{a}{5}}+\frac{a}{25}e^{\frac{a}{5}}\right)=-\frac{a}{25}e^{\frac{a}{5}}$ **A1**

For $a=5$, $\frac{d^2S}{da^2}<0$, hence local max. **R1**

Thus, $P=20-10+2e=10+2e$ **A1**

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[5 marks]